A Next-Generation AI Framework for Intelligent Travel Support in Sri Lanka

Progress Presentation 1 | 2026 January Batch





Presented By:

R26-IT-112

R26-IT-112

Comments from Proposal & Improvements

Student IT No	Comment Received	How We Addressed It
IT22607614	Nothing to do in this component. He is checking visa eligibility which can be done using some rule based decisions	The visa eligibility component was rejected due to the use of synthetic and non-real datasets, which reduced reliability for Sri Lankan visa decision scenarios. Therefore, the crowd prediction component was introduced as an alternative solution.
IT22216014	Emotion detection considering multiple aspects like location, day of the tour, past travellers experience, season, etc	The emotion forecasting feature was introduced to enhance traveller experience by analyzing factors such as location, tour day, season, and past traveller experiences to predict traveller emotions and provide personalized recommendations.
IT22082442	Need to come up with a new algorithm which incorprates optimization as well to deliver a dynamically changing tour plan considering budget duration weather, etc.	The proposed system utilizes the Non-dominated Sorting Genetic Algorithm II (NSGA-II) algorithm to optimize multiple factors such as budget, travel duration, weather conditions, and tourist preferences simultaneously. The algorithm dynamically generates and updates personalized tour itineraries in real time by selecting the most optimal travel plan from multiple possible solutions.
IT22176974	Component needs to be changed	This system uses a camera to detect driver drowsiness and sensors,speed to detect collision risks. Then it combines the data and calculates a risk score. Based on the risk level, alert messages are sent to the driver and passenger. It also has Face-ID verification and GPS-based road detection. The main novelty of this system is the AI-based risk scoring and real-time alert system and vibration system for better driver and passenger safety.

- Manual and inefficient travel planning
- Lack of real-time adaptive tourism systems
- Poor tourist experience management
- Tourist transportation safety risks
- Overcrowding at popular destinations



Problem

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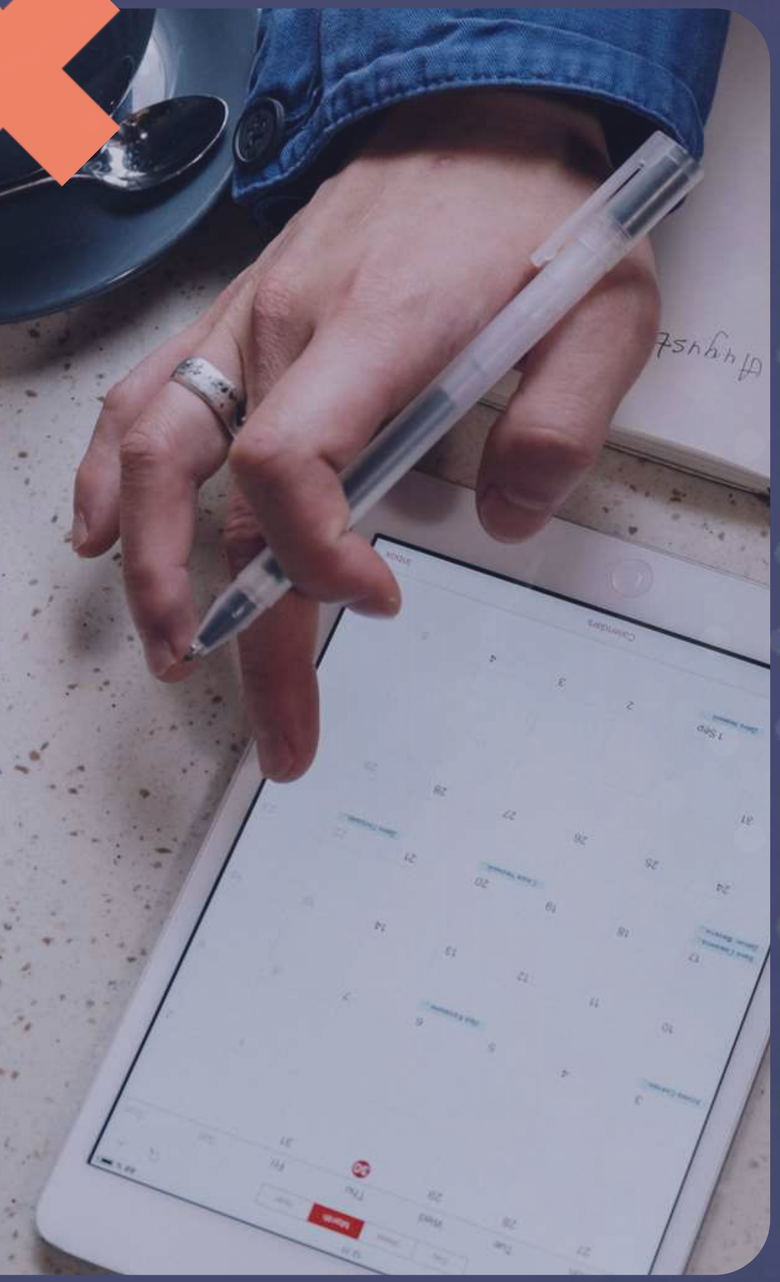
Impact / Gravity

- Tourists waste time and money
- Negative travel experiences reduce tourism satisfaction
- IIncreased accident risks during travel
- Poor crowd management affects destinations
- Underutilization of many tourist attractions
- Economic impact on Sri Lanka tourism sector

Beneficiaries



- International tourists
- Local tourists
- Travel agencies & Transport providers
- Tourism development authorities
- Hotel and hospitality industry
- Sri Lanka tourism economy



Design Excellence of the Overall Solution

01. 4 integrated AI-powered tourism components

02.Real-time adaptive intelligence

03.Predictive analytics & machine learning

04. NLP-based intelligent travel planning

05. Emotion-aware recommendation system

06. AI-driven tourist safety monitoring

07.GIS-based crowd optimization

08.Sri Lanka-specific smart tourism modeling

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IT22082442 | SENEVIRATHNA T.M.N.D.L

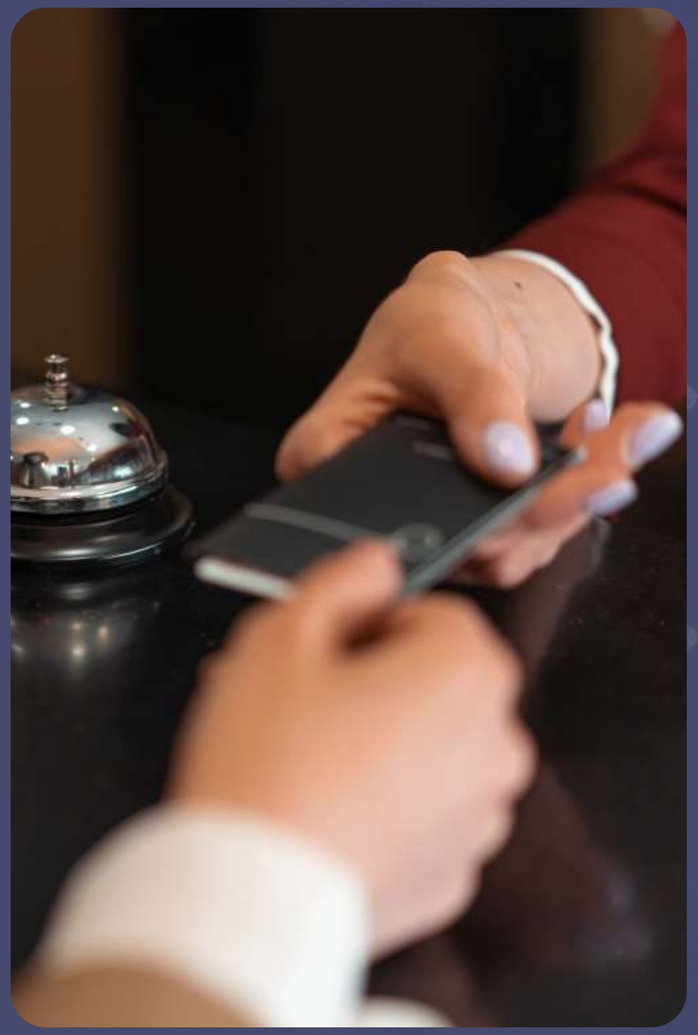
AI CONVERSATIONAL TRAVEL ANALYZER + INTELLIGENT PACKAGE GENERATOR + REAL TIME ADAPTIVE ENGINE



Specialization : Information Technology

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Sub-Problem / Prototype



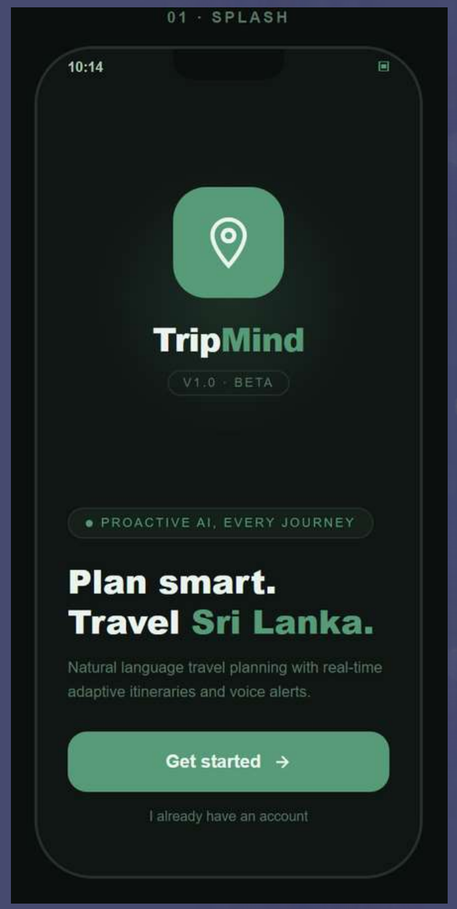
Tourists manually plan trips using multiple platforms, and existing systems cannot adapt to real-time disruptions such as weather changes, traffic congestion, or transport delays.

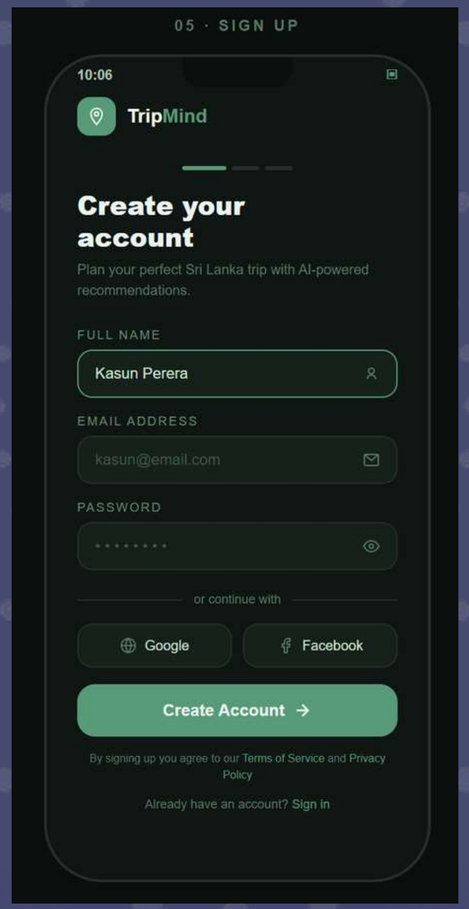
- Understand user travel requests using NLP
- Automatically generate optimized travel packages
- Monitor real-time travel conditions
- Dynamically update itineraries when disruptions occur
- Provide proactive voice-based notifications

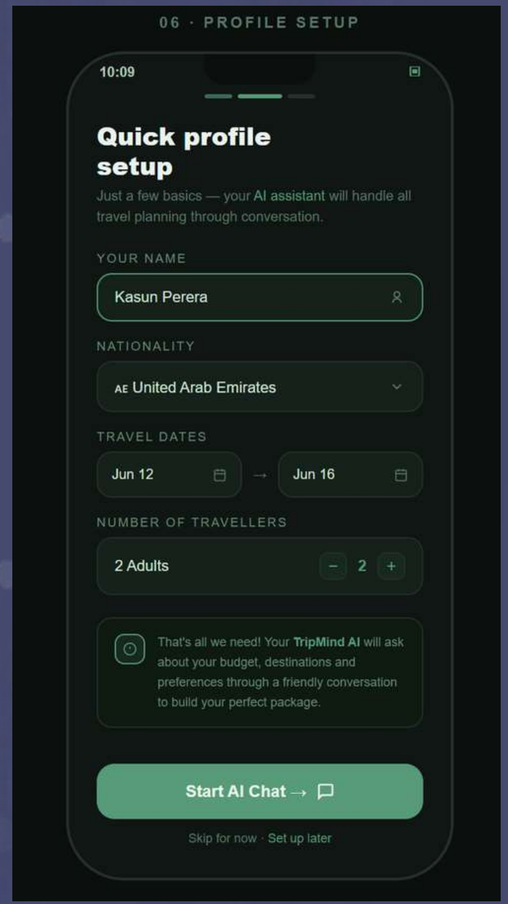


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Prototype Overview

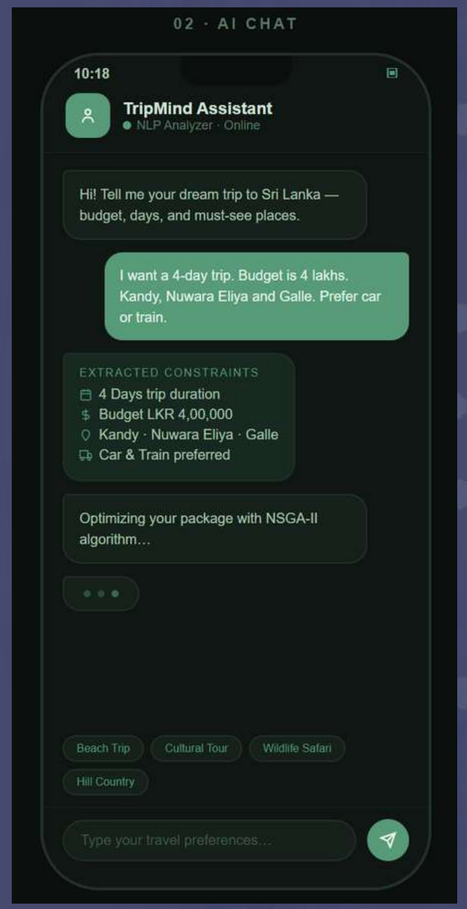


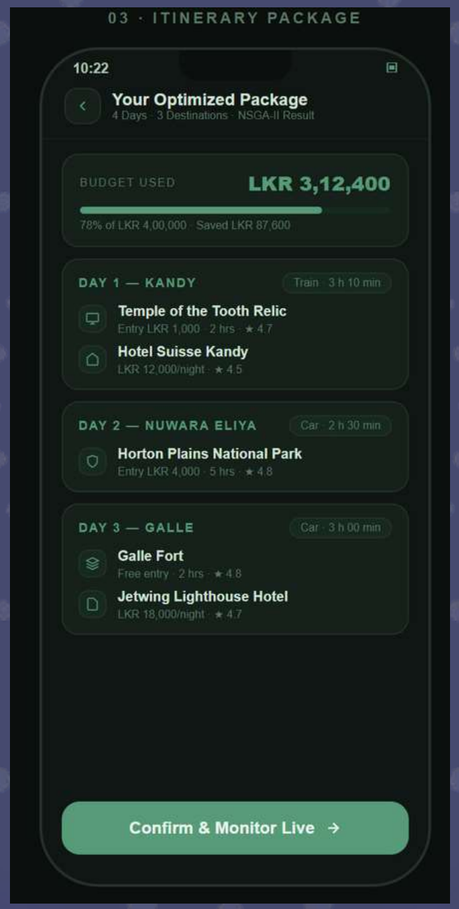


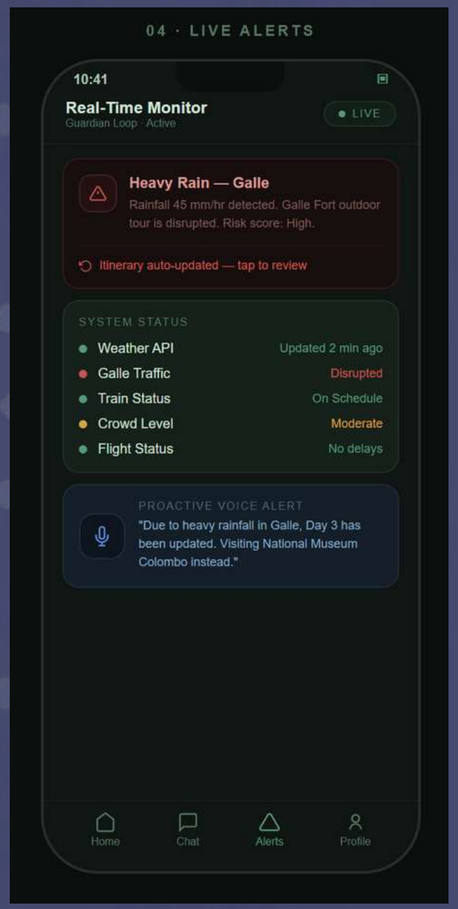


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Prototype Overview







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Prototype Overview



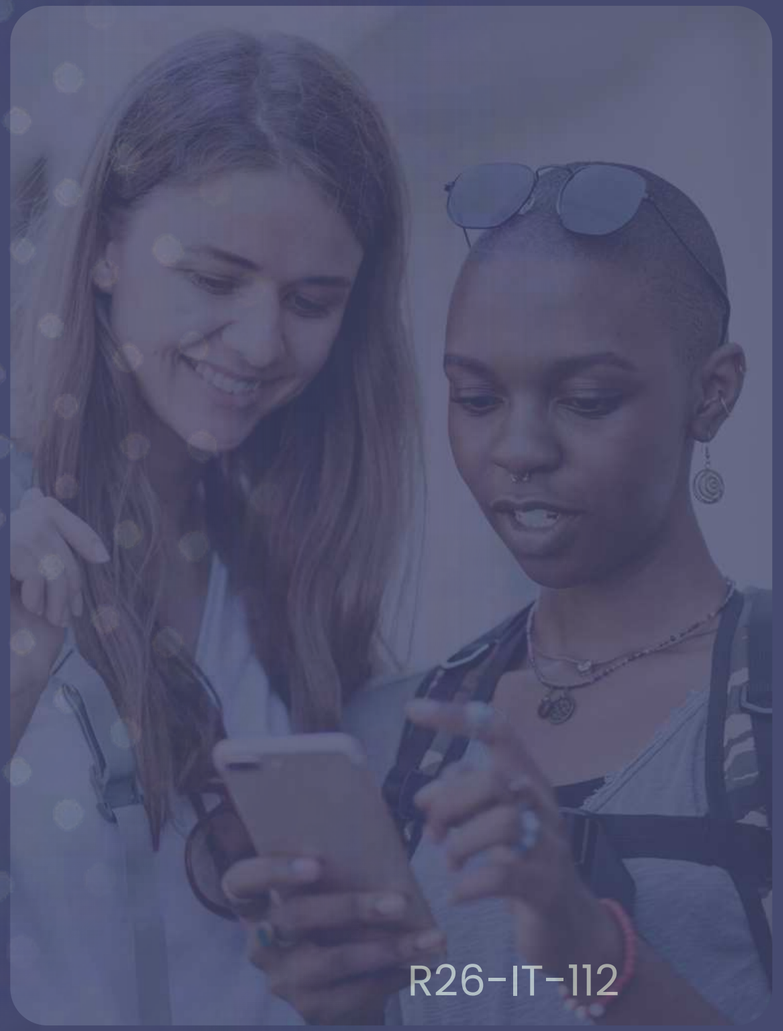




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User Requirements Addressed by the Solution

- Easy travel planning experience
- Personalized travel packages
- Budget optimization
- Real-time travel updates
- Reduced manual effort
- Faster decision-making



AI CONVERSATIONAL TRAVEL ANALYZER + INTELLIGENT PACKAGE GENERATOR + REAL TIME ADAPTIVE ENGINE

Design Excellence / Contribution

- Uses NLP for natural language travel understanding
- NSGA-II multi-objective optimization for itinerary generation
- Real-time adaptive travel engine
- Proactive AI recommendations instead of reactive systems
- Sri Lanka-specific tourism intelligence

Technical Excellence

- Llama 3 / GPT Models
- FastAPI
- React Native
- NSGA-II Algorithm
- Weather & Traffic APIs

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User Feedback on Prototype

- Users preferred automatic itinerary generation
- Real-time updates improved convenience
- Voice notifications were useful during disruptions
- Interface was easy to understand and use

Improvements Suggested by Users

- Add multilingual support



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IT22216014 | KAWEESHWARA M.D.O

ADAPTIVE EMOTION RECOGNITION FOR TOURIST EXPERIENCE MANAGEMENT

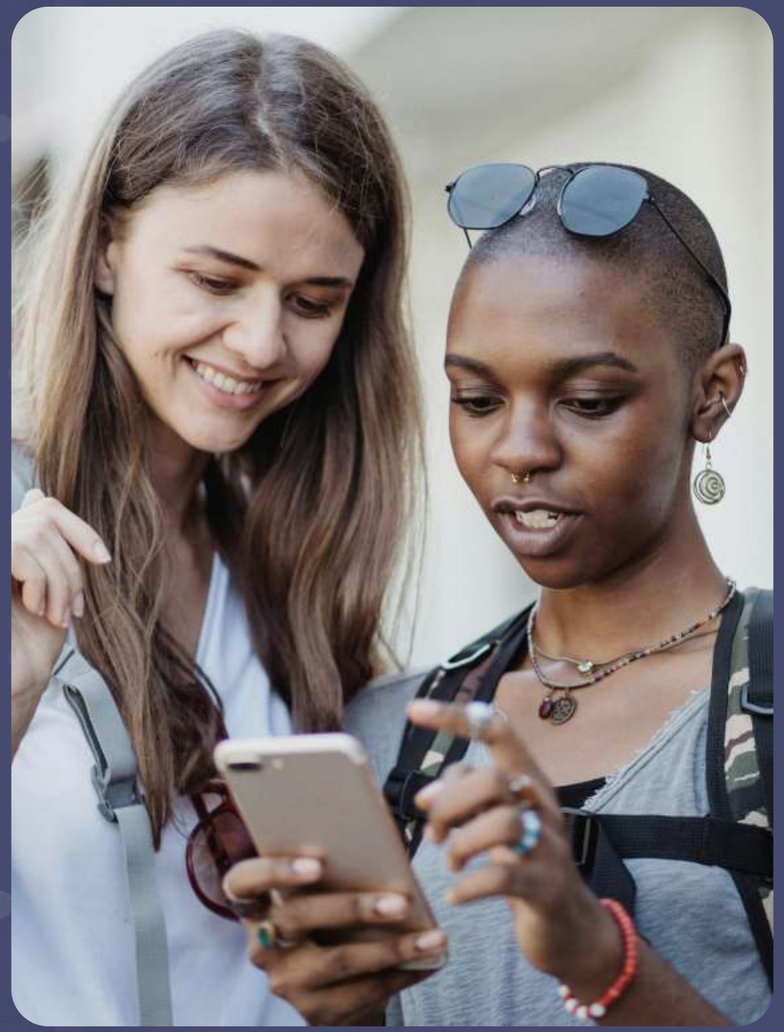
Specialization : Information Technology

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Sub-Problem / Prototype

ADAPTIVE EMOTION RECOGNITION FOR TOURIST EXPERIENCE MANAGEMENT

- Tourists struggle with real-time emotion understanding.
- Lack of adaptive travel experience support.
- No system predicts emotional changes using location, time, or history.



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Prototype Overview

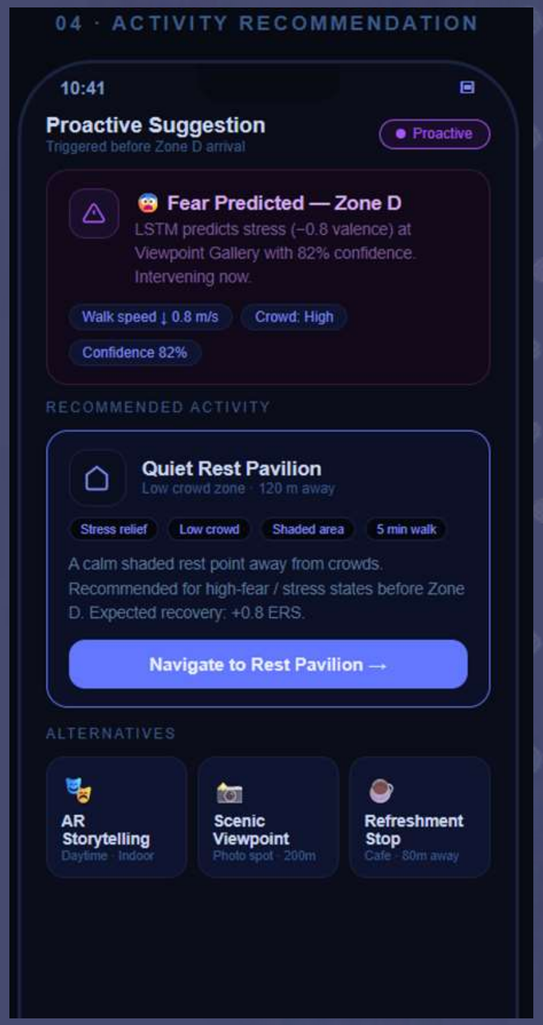






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Prototype Overview

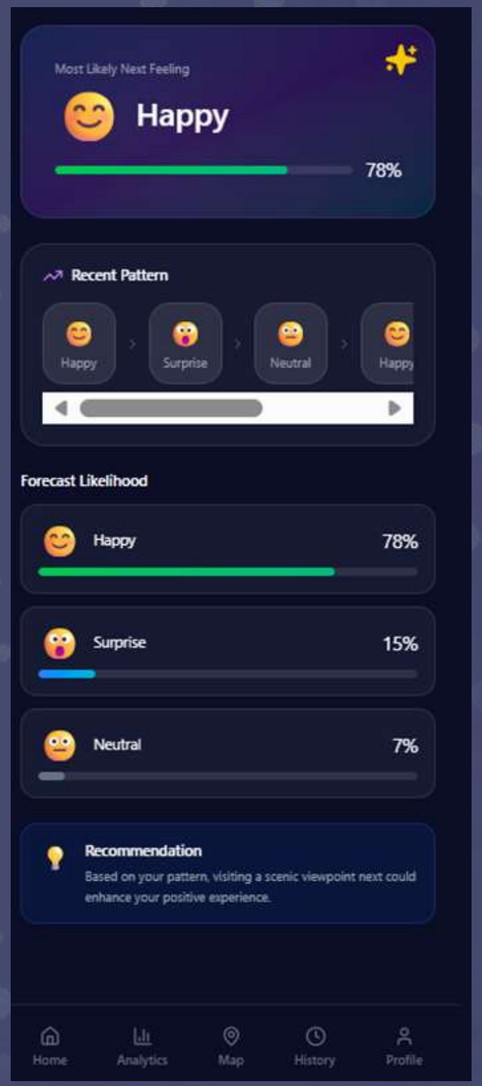




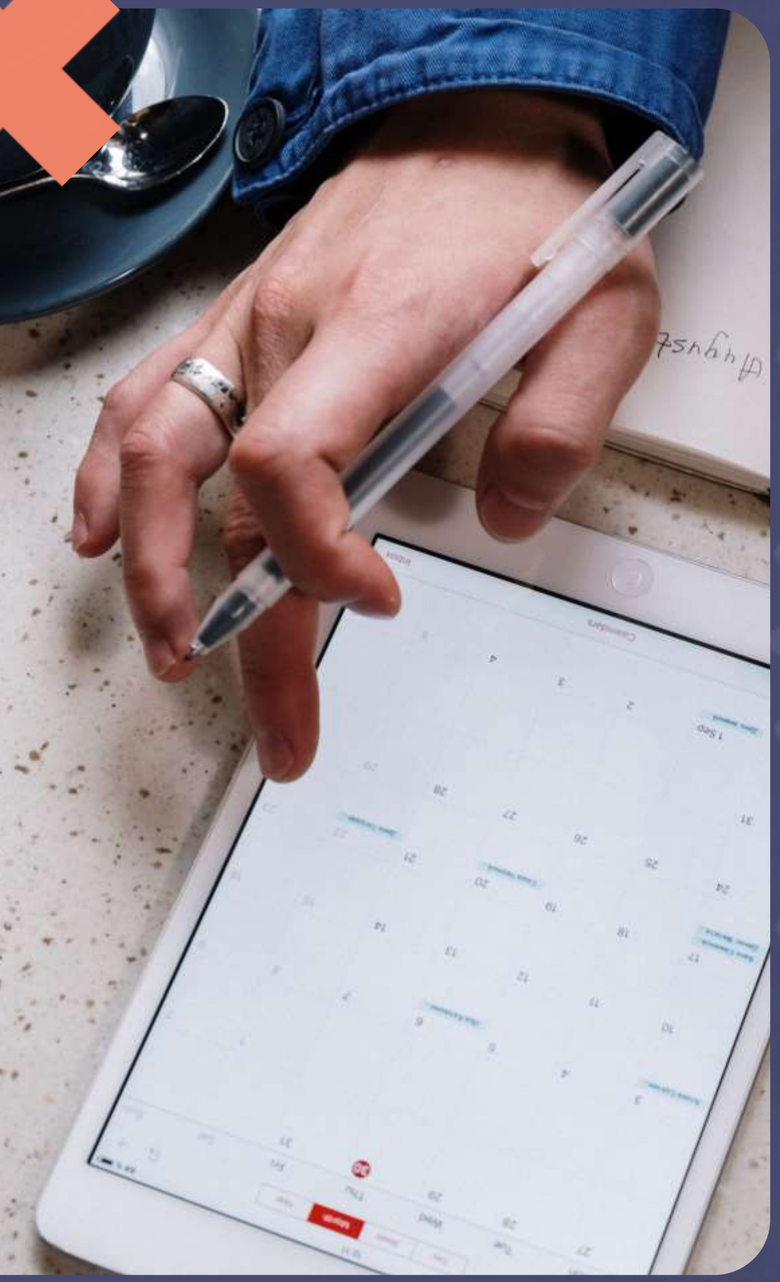
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Prototype Overview





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User Requirements Addressed by the Solution

01.Real-time emotion detection

02.Emotion tracking across locations

03.Improved tourist satisfaction

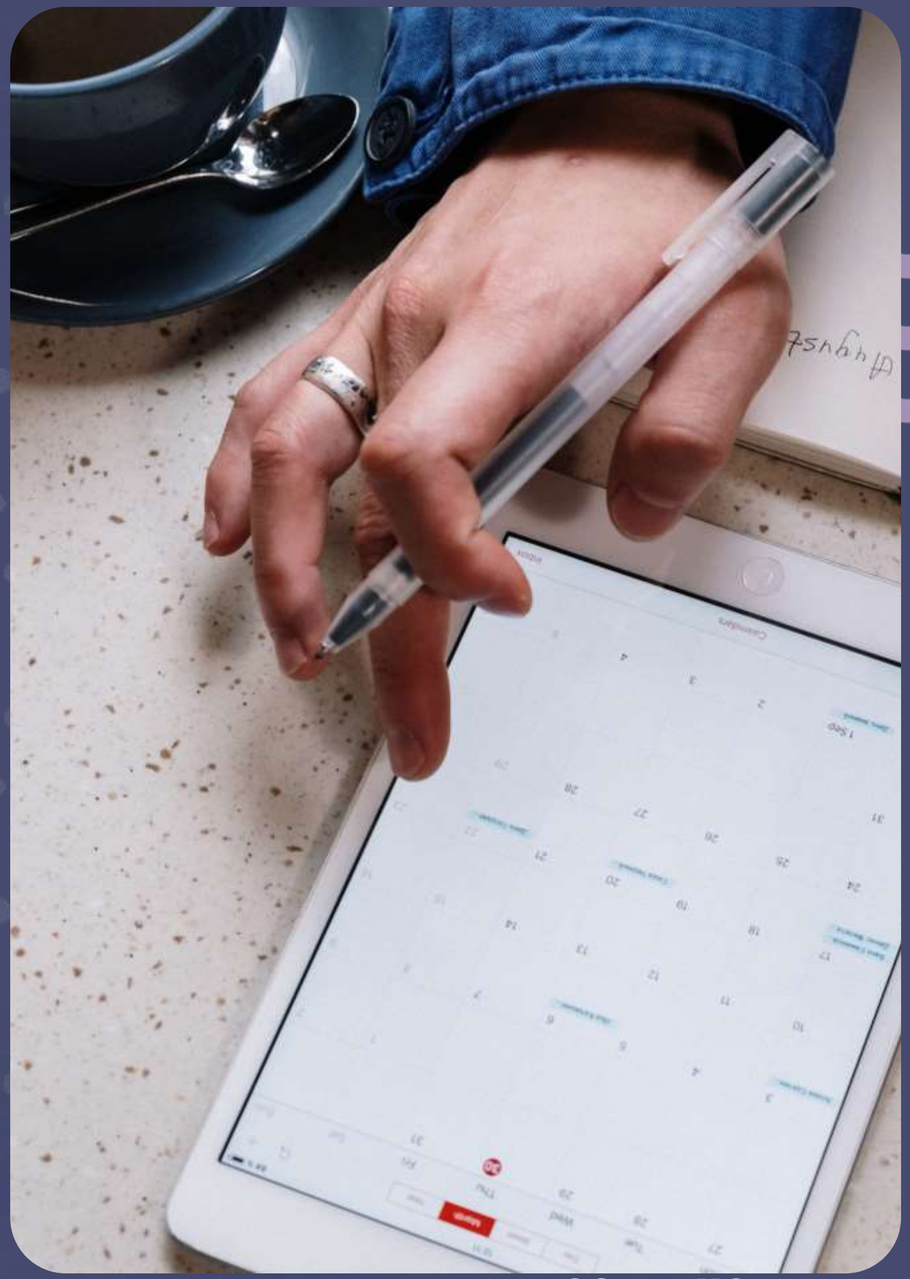
04.Predicts future emotional changes

05.Support for decision-making in travel experience management

ADAPTIVE EMOTION RECOGNITION FOR TOURIST EXPERIENCE MANAGEMENT

Technical Excellence

- MobileNetV2 transfer learning
- 7-class + 3-class emotion models
- Haarcascade face detection
- LSTM-based temporal emotion prediction
- Real-time face detection using OpenCV



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IT22176974 | RUPASINGHE M.L.K.S

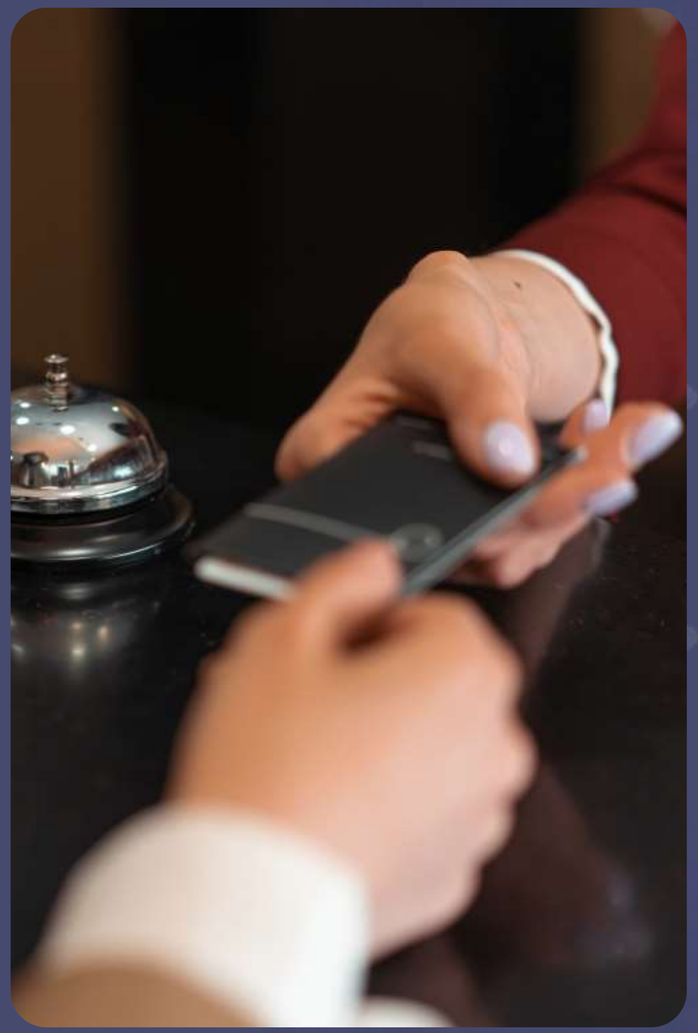


SMART DRIVER & VEHICLE PROACTIVE SAFETY SYSTEM

Specialization : Information Technology

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Sub-Problem / Prototype



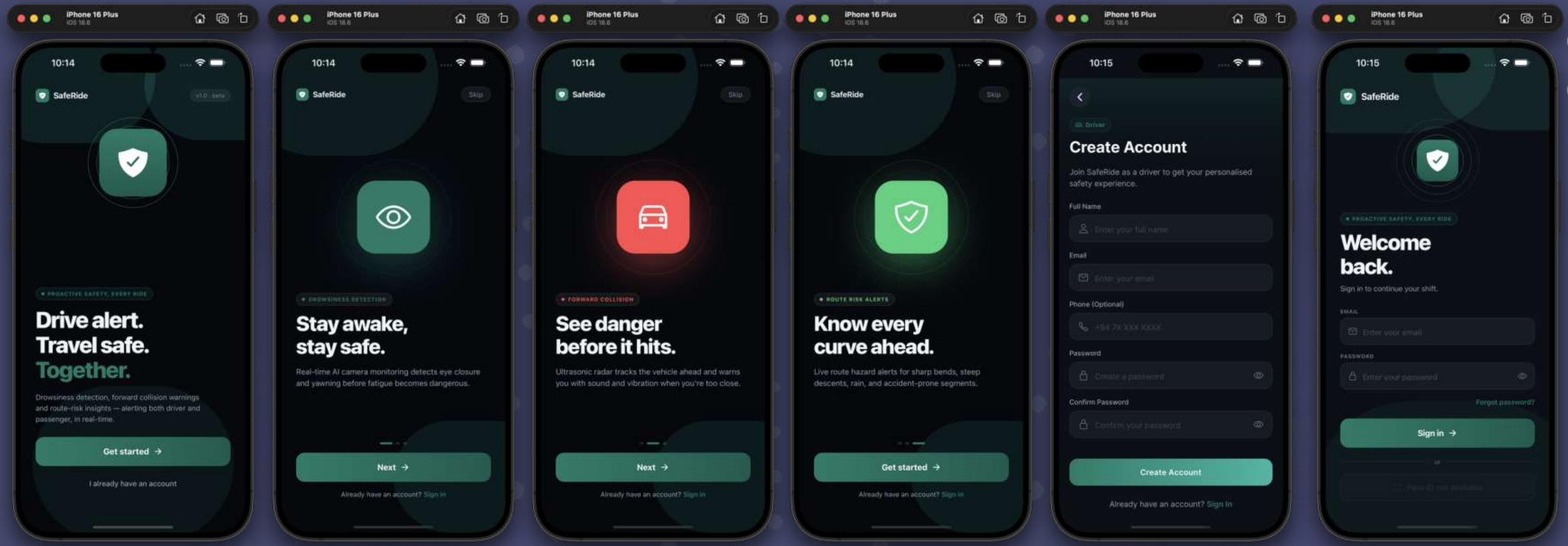
Addressing the critical gap in road safety through AI-driven driver monitoring and proactive alerts.

- IoT-based Driver Drowsiness/Fatigue Detection
- Forward Collision Warning using Radar Sensors
- Identify live location and speed using 4G GPS and speed sensor
- AI-based Combined Risk Scoring
- Simultaneous Real-time Alerts to Driver and Tourist, vibration systerm



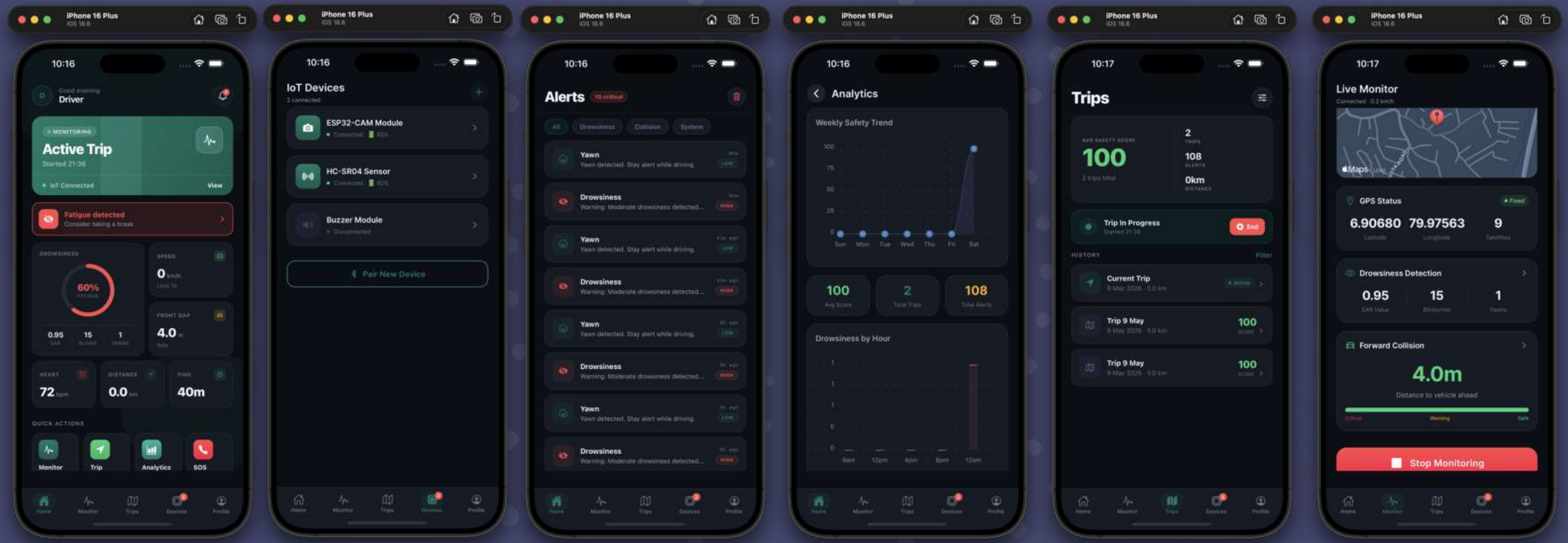
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Prototype Overview



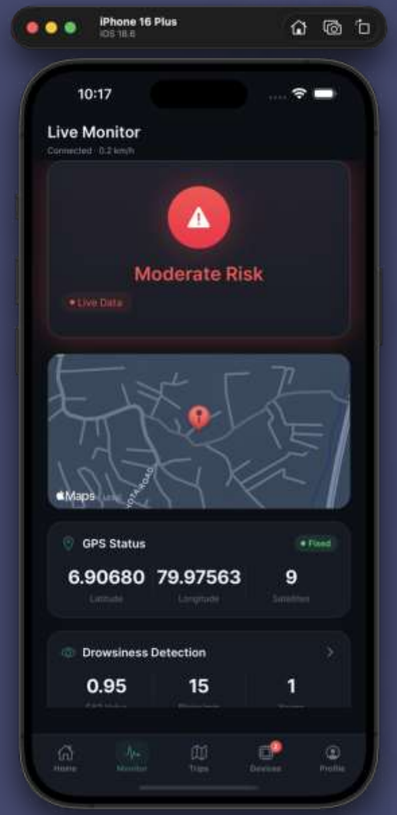
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Prototype Overview

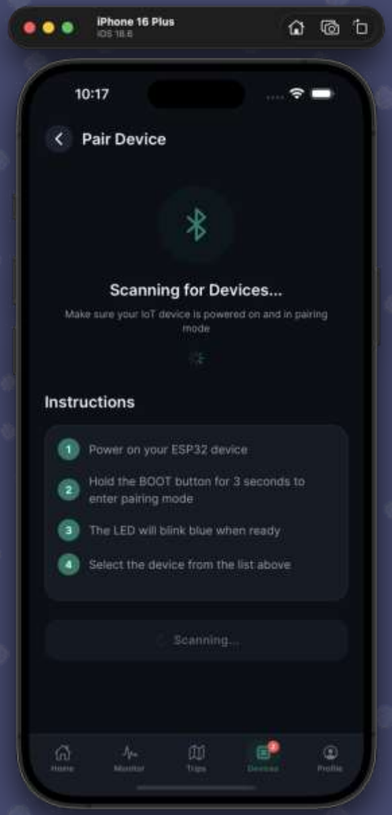


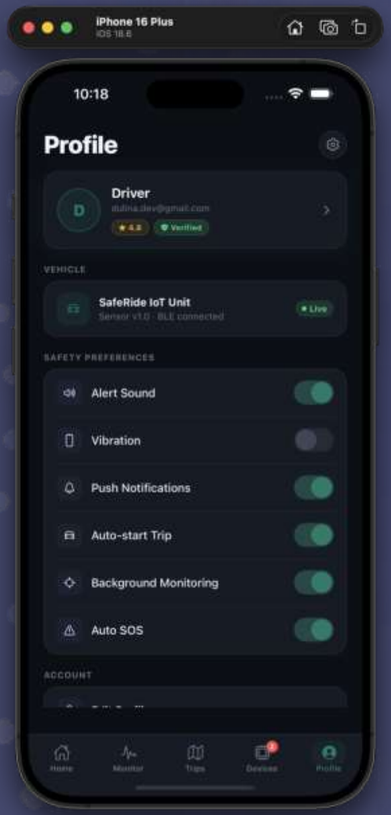
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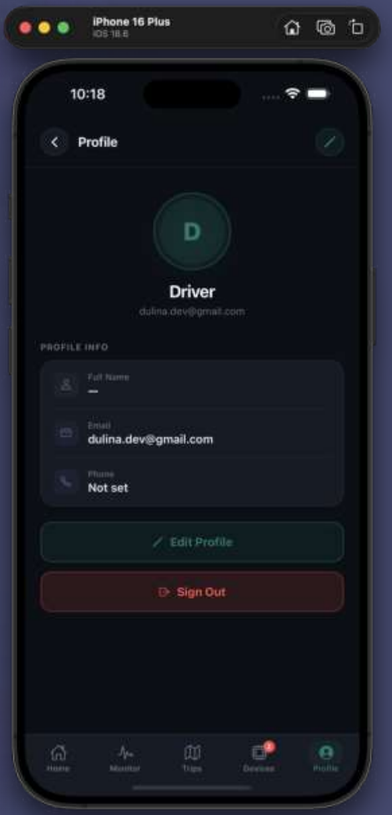
Prototype Overview







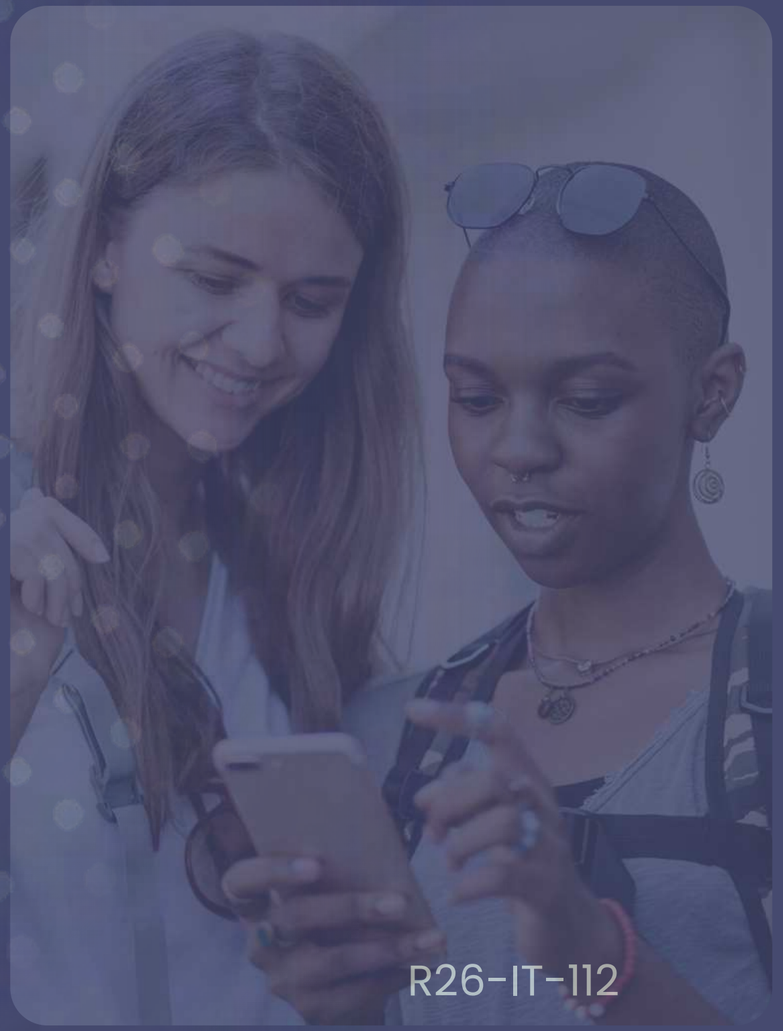




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User Requirements Addressed by the Solution

- Enhanced road safety experience
- Personalized safety alerts
- Real-time travel safety updates
- Reduced manual effort
- Faster decision-making



Design Excellence / Contribution

- Uses smart AI to catch small risks early before they turn into big accidents.
- Gives alerts to both the driver and the tourist so they can stay safe together.
- Works fast and reliably even at night or in areas with poor internet.
- Warns about dangerous road spots early by checking the vehicle's location.

Technical Excellence

- Facial/IoT camera for drowsiness (e.g., ESP32-CAM).
- Ultrasonic sensors for forward distance.
- GPS and speed sensor (4G)
- AI/ML for risk fusion
- IoT cloud for real-time alerts and data logging.
- •Mobile app for dual notifications (driver + tourist).

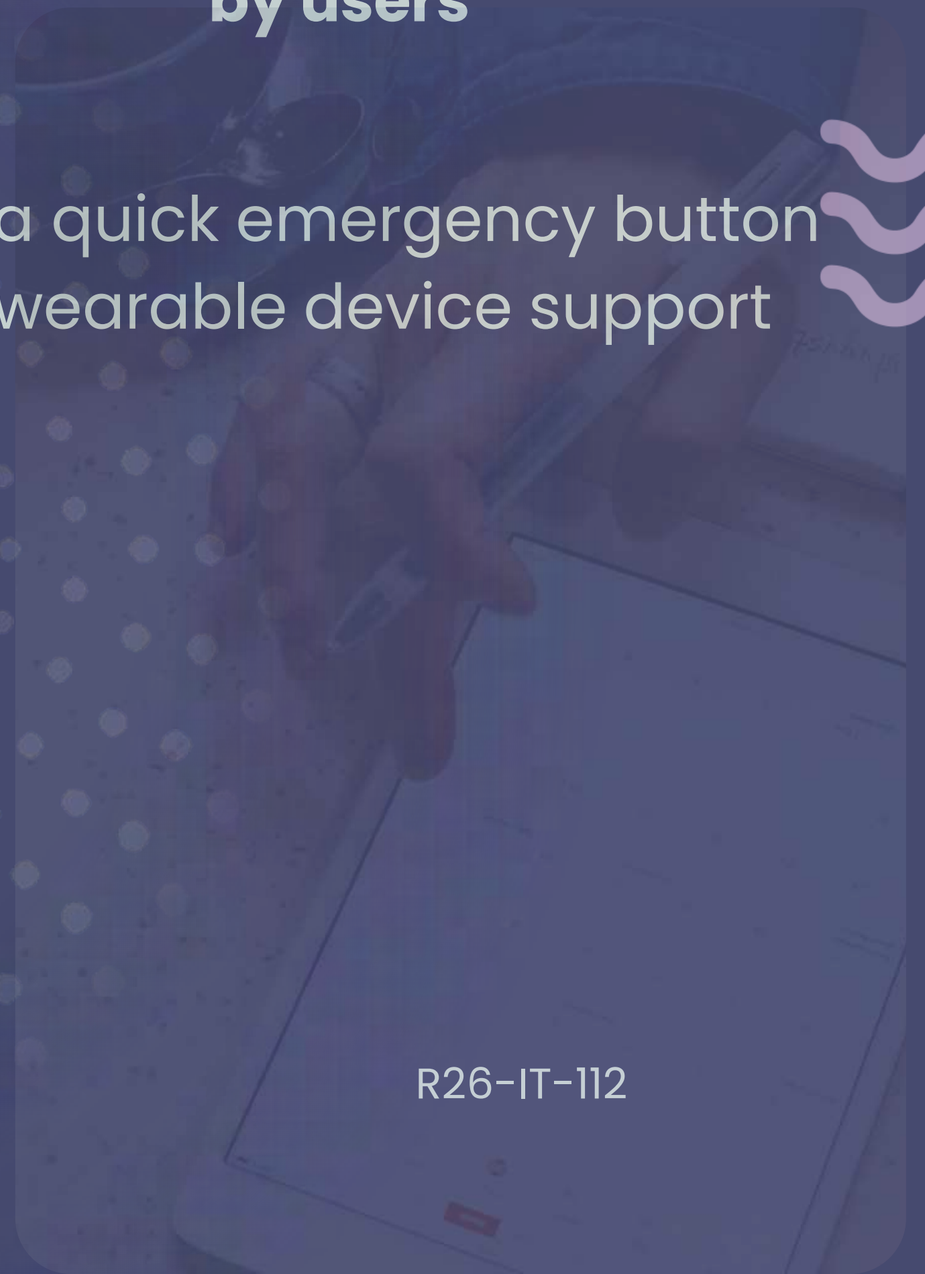
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User Feedback on Prototype

- Tourists feel much safer knowing they receive real time notifications about the driver's condition.
- Users appreciate the Face-ID login
- Users like the fast alerts because they work quickly even in dark places

Improvements Suggested by users

- Add a quick emergency button
- Add wearable device support



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IT22607614 | Manthrirathna M.A.A.C



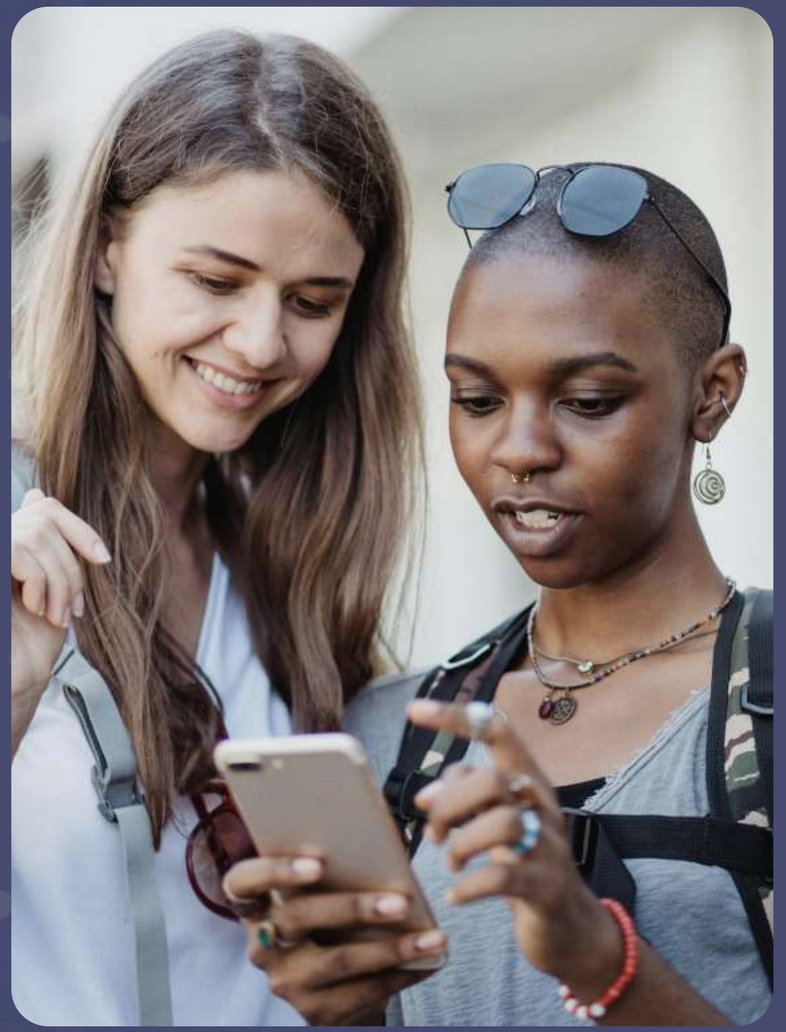
INTELLIGENT TOURISM FLOW OPTIMIZATION SYSTEM

Specialization : Information Technology

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INTELLIGENT TOURISM FLOW OPTIMIZATION SYSTEM

- Intelligent Tourism Flow Optimization System
- Reduces overcrowding in tourist destinations
- Uses AI and GIS-based analysis
- Improves tourist distribution efficiency



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KEY FEATURES

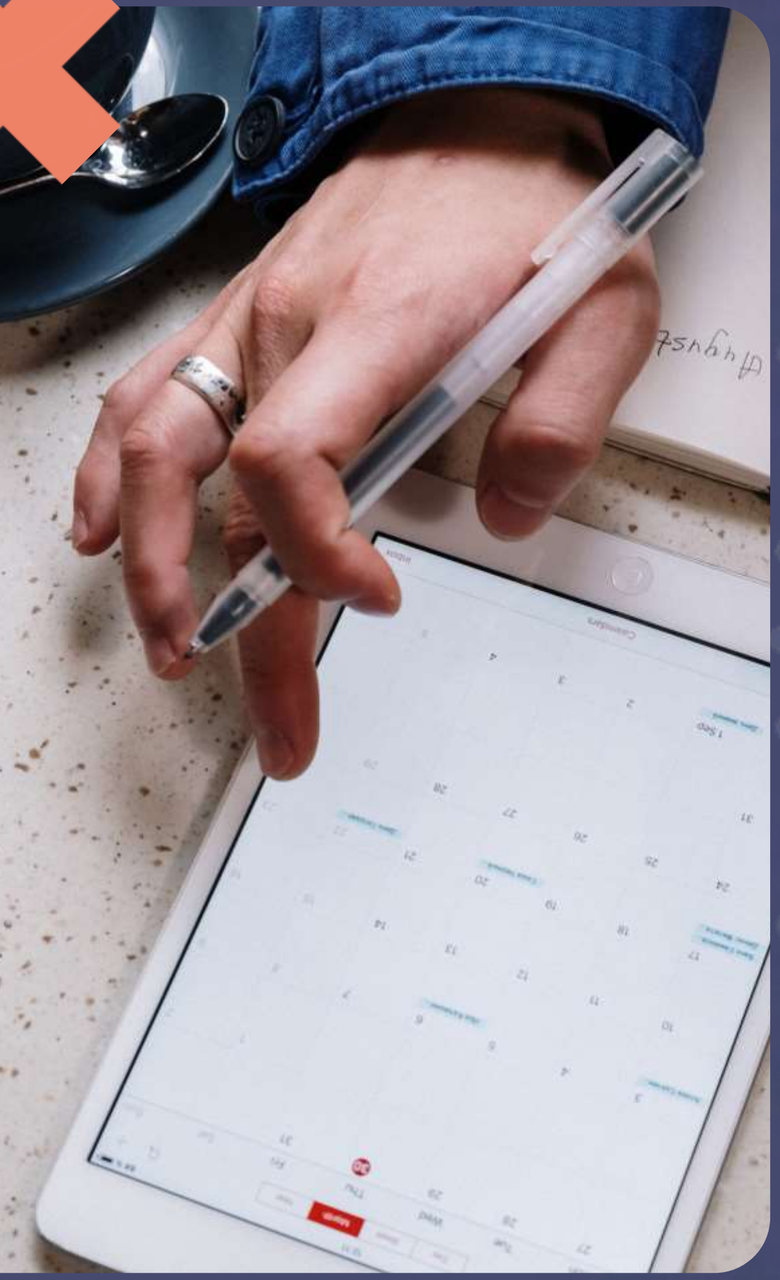
01.Real-time crowd monitoring system

02.Predictive crowd forecasting models

03.GIS-based heatmap visualization

04.Smart alternative destination suggestions and Seasonal tourism demand analysis

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SYSTEM WORKING

01.Collects real-time and historical tourism data

02.Processes data using machine learning models

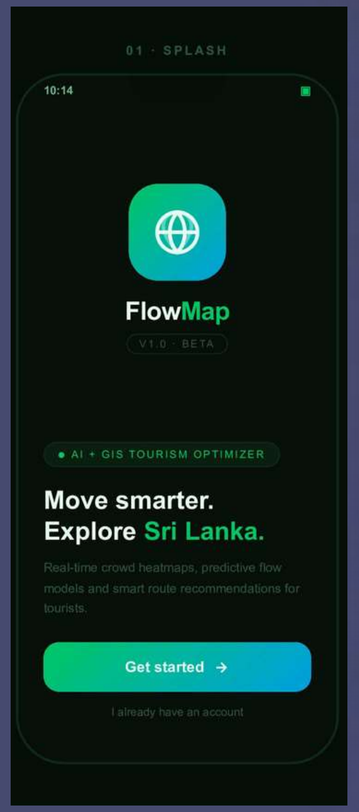
03. Predicts crowd density in locations

04. Generates heatmaps for visualization

05. Suggests less crowded alternative destinations

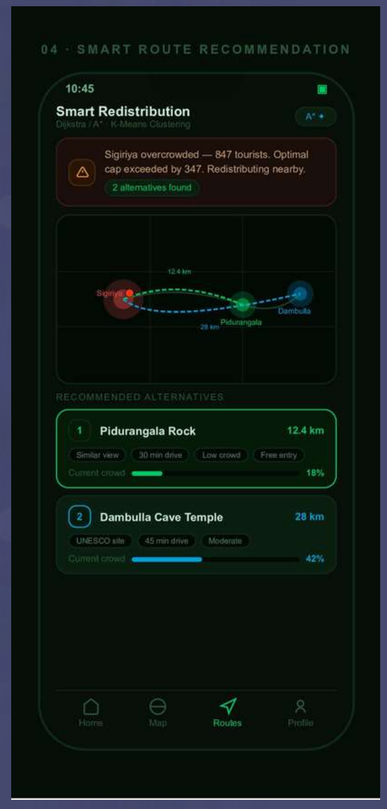
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Prototype Overview









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Thank You

